

TEST ONLY - NOT VALID FOR FIELD USE

EXCAVATION PERMIT AND SAFETY PLAN

Synthetic source document for Qoreva Step 2 document-intelligence testing

Document field	Test value	Document field	Test value
Document number	QTEST-EXC-001	Revision	0 - Test Draft
Project	Qoreva Test Project	Contractor	Test Civil Contractor
Work area	Building 6 - Storm Drain Alignment	Crew size	6
Planned work	Sep 12-19, 2026	Shift	Day shift
Permit status	NOT ISSUED - TEST DATA	Expiration	End of shift / condition change

TEST PURPOSE. This synthetic document contains realistic excavation controls, conditional requirements, named responsibilities, inspection points, and source references so Qoreva can be tested for document extraction, requirement matching, citations, and guided-question generation. It does not authorize work, replace a site-specific permit, or certify regulatory compliance.

1 Permit Scope and Work Limits

Work sequence	Planned activity	Authorized test boundary
1	Pre-task planning and utility verification	Review drawings, locates, work limits, permits, adjacent operations, and equipment assignments.
2	Expose and verify existing utilities	Daylight utilities by approved non-destructive methods before mechanical excavation enters the tolerance zone.
3	Excavate trench	Excavate to required line, grade, and depth using the selected protective system and competent-person oversight.
4	Install underground storm pipe	Lower, align, connect, and inspect pipe while controlling rigging, pinch points, suspended loads, and worker position.
5	Backfill, compact, and restore	Backfill in planned lifts, compact, remove temporary protection in sequence, and restore the area.

2 Permit Preconditions

Every item must be resolved before excavation begins; "N/A" requires a documented basis.

Status	Precondition / evidence required	Responsible role	Record / reference
☐ Yes ☐ No ☐ N/A	Approved PTP/JHA reviewed with the crew; scope and ordered work steps match field conditions.	Supervisor	PTP ID: _____
☐ Yes ☐ No ☐ N/A	Current public utility locate request is valid and field markings are legible.	Permit issuer	Ticket: _____ Expires: _____
☐ Yes ☐ No ☐ N/A	Private locating completed where owner records, private systems, lighting, or process utilities may exist.	Qualified locator	Report: _____
☐ Yes ☐ No ☐ N/A	Available drawings, records, owner information, and field markings have been compared.	Supervisor	Drawing/rev: _____
☐ Yes ☐ No ☐ N/A	Known or suspected utilities in the excavation influence zone are positively exposed or otherwise verified.	Competent person	Pothole log: _____
☐ Yes ☐ No ☐ N/A	A competent person has inspected the excavation area and has authority to stop work and correct hazards.	Competent person	Inspection time: _____
☐ Yes ☐ No ☐ N/A	Protective system, access/egress, spoil placement, traffic control, and water controls are available before entry.	Competent person	System/data: _____

3 Utility Location, Exposure, and Protection Requirements

- Obtain applicable utility locate information before disturbing the ground. The crew shall not rely on drawings or surface markings alone when the actual horizontal location or depth remains uncertain.
- Review available drawings, owner records, locator reports, and field markings during the pre-task briefing. Document discrepancies before work begins.
- Use private locating or additional locating methods when project conditions indicate private, abandoned, lighting, communication, process, or undocumented systems may be present.
- Positively expose or otherwise verify utilities before mechanical excavation can encroach on the project-defined tolerance zone. Use approved non-destructive methods appropriate to the utility and soil conditions.
- Stop mechanical excavation immediately when a utility cannot be verified, a mark is missing, a facility is found outside the expected location, or field conditions conflict with the plan.
- Maintain the owner-required clearance from exposed utilities. Support exposed facilities before removing soil support and prevent movement, deflection, separation, or loading damage.
- If an energized or pressurized service is contacted, damaged, or suspected of damage: stop work, keep personnel clear, prevent ignition or secondary contact, call emergency services when needed, and notify the owner/operator.

Utility / system	Locate source	Field mark	Positive exposure	Status / notes
Electrical / site lighting	Public + private locate	Red	☐ Required ☐ Complete	_____
Natural gas / fuel	Public + owner record	Yellow	☐ Required ☐ Complete	_____
Water / fire protection	Public + owner record	Blue	☐ Required ☐ Complete	_____
Storm / sanitary sewer	Survey + owner record	Green	☐ Required ☐ Complete	_____

Utility / system	Locate source	Field mark	Positive exposure	Status / notes
Communication / fiber	Public + private locate	Orange	☐ Required ☐ Complete	_____
Unknown / abandoned systems	Visual + locator sweep	White/pink notes	☐ Required ☐ Complete	_____

4 Competent Person, Soil, and Protective System

Decision point	Selection / record
Competent person	Name: _____ Employer: _____ Contact: _____
Maximum anticipated depth	_____ ft Maximum width: _____ ft Length: _____ ft
Soil classification	☐ Stable rock ☐ Type A ☐ Type B ☐ Type C ☐ Layered/mixed ☐ Not yet determined
Protective system	☐ Sloping ☐ Benching ☐ Shoring ☐ Shield/trench box ☐ Engineered system ☐ No entry
Design basis	☐ 29 CFR 1926 Subpart P ☐ Manufacturer tabulated data ☐ RPE design ☐ Other: _____
Water condition	☐ Dry ☐ Seepage ☐ Accumulating water ☐ Dewatering required
Adjacent loading	☐ Spoil ☐ Mobile equipment ☐ Structures ☐ Traffic ☐ Vibration ☐ None identified

- Protect each employee in an excavation from cave-in hazards with an adequate protective system except when the excavation is entirely in stable rock or is less than 5 feet deep and a competent person finds no indication of a potential cave-in.
- A protective system for an excavation 20 feet deep or greater must be designed by a registered professional engineer or based on tabulated data prepared or approved by one.
- The competent person shall classify soil and select the protective system using actual depth, soil, water, weather, vibration, surcharge loading, nearby structures, and adjacent operations.
- Employees may not work outside a trench shield or other designated protected area where a cave-in exposure exists.

5 Hazard-to-Control Plan

Each credible hazard is paired with task-specific controls and a verification method.

ID	Hazard	Required controls	Verification
H1	Cave-in or soil collapse	CP inspection; soil classification; sloping, benching, shoring, or shielding; protect against surcharge and changing conditions.	CP records system and inspects before entry.
H2	Contact with underground utilities	Current locates; records review; private locate when indicated; positive exposure; clearance; stop on uncertainty.	Locate ticket and pothole log verified.
H3	Worker struck by mobile equipment	Defined travel paths; barricaded swing radius; spotter where visibility is limited; high-visibility apparel; communication plan.	Supervisor observes separation and signals.
H4	Falling material or edge surcharge	Keep spoil, materials, and equipment at least 2 feet from the edge or use retaining devices sufficient to prevent material entry.	CP measures setback and checks retention.

ID	Hazard	Required controls	Verification
H5	Unsafe access or egress	For trenches 4 feet or deeper, provide stairway, ladder, ramp, or other safe egress so no worker travels more than 25 lateral feet.	CP measures access spacing and secures ladders.
H6	Water accumulation / instability	Prevent entry where water is accumulating unless adequate precautions are used; provide monitored dewatering and inspect after water intrusion.	Pump capacity and discharge route checked.
H7	Hazardous atmosphere	Where a hazardous atmosphere could reasonably exist, test before entry; ventilate or protect workers; continue monitoring as conditions require.	Calibrated instrument and test log reviewed.
H8	Suspended loads / pipe handling	Use inspected rigging; qualified operator and signal person; tag lines when appropriate; no employee beneath suspended loads.	Pre-lift inspection and exclusion zone confirmed.
H9	Falls into excavation	Barricades, guardrails, covers, warning lines, or controlled access appropriate to the edge and public/worker exposure.	Supervisor checks perimeter before shift.
H10	Adjacent structures or utilities lose support	Evaluate influence zone; underpin, brace, support, or obtain engineered direction before removing support.	CP/RPE documents support method.

6 Access, Material Placement, and Equipment Controls

- Provide safe access and egress for trench excavations 4 feet or deeper. Ladders, steps, ramps, or other safe means of exit shall be positioned within 25 lateral feet of workers and kept inside the protected area where applicable.
- Keep spoil piles, materials, and other surcharge loads at least 2 feet from the excavation edge, or use retaining devices sufficient to prevent material from falling or rolling into the excavation.
- When mobile equipment approaches an excavation and the operator lacks a clear direct view of the edge, use a warning system such as barricades, hand/mechanical signals, stop logs, or another effective method.
- Keep employees clear of loads handled by lifting or digging equipment. Employees may not stand beneath suspended loads or in a location exposed to a falling load.
- Use a spotter when backing, congestion, blind spots, swing radius, overhead/underground utilities, or adjacent work creates an exposure that the operator cannot control alone.

Control	Required value / selection	Verified by
Spoil setback	Minimum 2 ft or effective retaining device	_____
Egress spacing	Maximum 25 lateral ft travel in trench 4 ft or deeper	_____
Edge warning system	☐ Barricade ☐ Stop log ☐ Spotter ☐ Signal ☐ Other: _____	_____
Equipment exclusion zone	Swing radius and travel path physically defined	_____
Pipe lowering method	Rigging ID / capacity: _____	_____

7 Atmosphere, Water, Weather, and Changing Conditions

Condition	Required action	Record
Hazardous atmosphere could reasonably exist	Test before employee entry into excavations deeper than 4 feet. Provide ventilation, respiratory protection, rescue equipment, and continued monitoring as required by the exposure.	O2: ____% LEL: ____% H2S: ____ ppm CO: ____ ppm
Water accumulating or flowing into excavation	Do not permit entry unless adequate precautions protect employees. Use pumps, drainage, special support/shielding, or a competent-person approved combination.	Method: _____
Rain, thaw, freeze, vibration, nearby traffic, or loading change	Stop entry and require a competent-person reinspection before work resumes.	Reinspection: ____ Date/time: ____
Evidence of cave-in, protective-system distress, hazardous atmosphere, or other hazardous condition	Remove exposed employees immediately until precautions are taken and the condition is corrected.	Corrective action: _____

8 Task-Specific PPE and Emergency Readiness

Category	Minimum / conditional requirement
Minimum PPE	Hard hat; safety glasses with side shields; high-visibility apparel; work boots; task-appropriate gloves.
Conditional PPE	Hearing protection, face shield, chemical-resistant gloves/clothing, respiratory protection, fall protection, or arc-rated protection when the hazard assessment requires it.
Emergency communication	Charged phone/radio available; emergency address and access route communicated to the crew.
Rescue limitation	No unplanned entry into a collapsed excavation or hazardous atmosphere. Call emergency services and use the site emergency plan.
Utility incident	Keep clear; control ignition/contact hazards; notify emergency services and utility owner/operator; preserve the scene when safe.
Medical response	First-aid kit and trained responder available as required; nearest treatment facility and route identified.

Emergency planning field	Test entry / field verification
Emergency address / location	Building 6 - Qoreva Test Project (confirm actual address before any real work)
Site emergency number	911 / Site number: _____
Nearest medical facility	Facility: _____ Route: _____
Assembly point	_____
Utility emergency contacts	Electrical: _____ Gas: _____ Water: _____ Owner: _____

9 Stop-Work and Reassessment Triggers

- Utility location, depth, identity, or status cannot be verified; markings are missing, damaged, or inconsistent with field conditions.
- Soil, depth, water, weather, vibration, surcharge, adjacent structure, or access conditions differ from the approved plan.

- Protective system is damaged, improperly installed, outside manufacturer/tabulated-data limits, or no longer appropriate.
- A hazardous atmosphere is detected, atmospheric monitoring fails, or required rescue capability is unavailable.
- Water accumulates, soil sloughs, tension cracks appear, material falls, or any sign of instability develops.
- The crew, competent person, operator, spotter, or supervisor identifies a new or uncontrolled exposure.
- Scope, sequence, crew, equipment, utility status, or adjacent work changes materially. Revise the PTP/permit before resuming.

10 Competent-Person Inspection Log

Inspect before work starts, daily before entry, and after rainstorms or other hazard-increasing events.

Inspection item	Acceptable	Deficiency / corrective action
Surface encumbrances and adjacent structures addressed	☐ Yes ☐ No ☐ N/A	_____
Utilities located, marked, and positively verified as required	☐ Yes ☐ No ☐ N/A	_____
Soil classification and protective-system selection remain valid	☐ Yes ☐ No ☐ N/A	_____
Protective system installed per design/tabulated data	☐ Yes ☐ No ☐ N/A	_____
Spoil/material/equipment setback or retention is effective	☐ Yes ☐ No ☐ N/A	_____
Safe access/egress is installed, secured, and within required travel distance	☐ Yes ☐ No ☐ N/A	_____
Water controls are adequate; no hazardous accumulation	☐ Yes ☐ No ☐ N/A	_____
Atmospheric test completed where a hazardous atmosphere could exist	☐ Yes ☐ No ☐ N/A	_____
Barricades, fall protection, traffic control, and equipment separation are effective	☐ Yes ☐ No ☐ N/A	_____
No employees exposed to suspended loads or unprotected cave-in areas	☐ Yes ☐ No ☐ N/A	_____
Conditions match the PTP, permit, and crew briefing	☐ Yes ☐ No ☐ N/A	_____

Inspection event	Date / time	Competent person	Result
Initial / pre-entry	_____	_____	☐ Acceptable ☐ Stop work
Start of shift	_____	_____	☐ Acceptable ☐ Stop work
After rain / water intrusion	_____	_____	☐ Acceptable ☐ Stop work
After vibration / loading / condition change	_____	_____	☐ Acceptable ☐ Stop work

11 Crew Briefing and Authorization Record

By signing, each test participant acknowledges the planned scope, hazards, controls, emergency actions, stop-work authority, and requirement to report changed conditions. These blank signature lines do not constitute approval and must not be used for field authorization.

Crew member	Role	Signature	Date / time
_____	_____	_____	_____

Crew member	Role	Signature	Date / time
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

Role	Printed Name	Signature	Date / Time
Permit issuer	Name: _____	Signature: _____	Date/Time: _____
Competent person	Name: _____	Signature: _____	Date/Time: _____
Field supervisor	Name: _____	Signature: _____	Date/Time: _____
Safety reviewer	Name: _____	Signature: _____	Date/Time: _____

12 Permit Validity, Suspension, and Closeout

- A real excavation permit must be issued by the authorized employer/site representative before work begins. This synthetic test document is never valid for field use.
- A permit is suspended when its expiration is reached; the shift changes; the competent person is no longer available; or scope, weather, soil, utilities, depth, protective system, equipment, crew, or adjacent operations change materially.
- Before resuming, the competent person and permit issuer must reassess conditions, correct deficiencies, update the PTP/permit, and rebrief affected workers.
- At closeout, remove personnel and tools, confirm utilities and installed work are supported, remove protective systems in a controlled sequence, backfill/compact as planned, restore barricades or surfaces, and document unresolved conditions.

Closeout item	Status / record
Employees, tools, rigging, and temporary services removed	☐ Complete ☐ N/A Notes: _____
Installed utilities and exposed existing utilities supported/protected	☐ Complete ☐ N/A Notes: _____
Protective system removed in planned sequence	☐ Complete ☐ N/A Notes: _____
Backfill, compaction, and surface restoration completed	☐ Complete ☐ N/A Notes: _____
Open excavation secured if work is incomplete	☐ Complete ☐ N/A Notes: _____
Permit closed by authorized person	Name: _____ Signature: _____ Date/time: _____

13 Source References and Extraction Test Targets

Source / label	What Qoreva should be able to surface
29 CFR 1926 Subpart P - Excavations	Applicable excavation requirements, competent-person duties, protective systems, access/egress, atmosphere, water, and inspections.
OSHA Construction eTool - Trenching and Excavation	Practical trenching guidance, safe egress, spoil placement, protective systems, and competent-person inspection.
OSHA Trenching and Excavation Safety Fact Sheet (OSHA 3267)	5-foot protective-system threshold, 20-foot engineered-system threshold, 4-foot egress trigger, 25-foot travel distance, and 2-foot spoil setback.
Project rule - QTEST-EXC-001	Positive exposure, stop-work triggers, utility clearance, equipment separation, PPE, emergency response, verification, and authorization fields.

Official reference links (accessed September 12, 2026): <https://www.osha.gov/etools/construction/trenching> | <https://www.osha.gov/trenching-excavation>

14 Qoreva Upload Test Checklist

Use this page to record what Step 2 actually detects from the document.

Expected extraction target	Observed result after upload	Pass
Document type identified as an excavation permit / safety plan	_____	☐
Project, contractor, work area, crew size, dates, and document revision	_____	☐
Applicable work activities and ordered five-step sequence	_____	☐
Utility locate, private locate, positive exposure, and stop-work requirements	_____	☐
Competent-person inspection and soil/protective-system decisions	_____	☐
5-ft protective-system, 20-ft engineered design, 4-ft egress, 25-ft travel, and 2-ft spoil thresholds	_____	☐
Hazard-to-control relationships H1 through H10 with verification methods	_____	☐
Atmosphere, water, equipment, suspended-load, PPE, and emergency requirements	_____	☐
Stop-work triggers and permit suspension / reissue conditions	_____	☐
Source labels or citations tied to the extracted requirement text	_____	☐

Recommended test: Attach the PDF in Step 2, allow processing to finish, then inspect detected requirements, source locations, applicable activities, and any new Guided Planning questions. Record missing, duplicated, or incorrectly scoped findings above before changing the source document or extraction logic.

END OF SYNTHETIC TEST DOCUMENT - NO FIELD AUTHORIZATION